



Design-based stereology in alpaca skin (*Vicugna pacos*): Impacts on fiber production

Tais Harumi de Castro Sasahara^a, Concepta Margaret McManus^b,
Luciana Milagros Durand Neira^a, Silvio Pires Gomes^a, Jessica Borghesi^a,
Marcella Giancoli - Kato^a, Alice Marcela Sampaio Del Colletto^a, Ana Claudia Oliveira Carreira^a,
Guilherme Chiarello^a, Maria Angelica Miglino^{a,*}

^a Department of Surgery, School of Veterinary Medicine and Animal Science, University of São Paulo, São Paulo, Brazil

^b Instituto de Ciências Biológicas, Campus Darcy Ribeiro, Universidade de Brasília, Brasília, DF, 79910-900, Brazil

ARTICLE INFO

Keywords:

South American camelid

Histology

Integument

Stereology

ABSTRACT

Alpacas are reared mainly for fiber extraction, which is a highly valued product in the textile industry. The quality of fiber has close relation to the skin. For this reason, this work aims to evaluate the skin of different body regions; scapular, costal and lateral femoral in Huacaya and Suri alpacas by means of stereology, an accurate quantification method. The total volume of the skin, derma, follicular groups and sebaceous and sweat glands was estimated. Total volume of skin (Vref) and total volume of follicular groups (VGF) were different for body regions. Femoral region showed higher values for VGF. Colour and breed were also different for Vref and total volume of derma (VD). Comparing the two breeds the femoral region presented higher fiber production. Dark animals had more derma and it was reported close relationship between total skin volume and their fractions volumes: derma, follicular groups and sweat glands.

1. Introduction

Alpacas (*Vicugna pacos*) are a group of South American camelids, found in the upper Andean regions of South America. There are two varieties: the Huacaya and Suri. Alpaca fleece is one of the most valuable animal products in the textile industry and has a wide range of colors (Gutiérrez Hermosa, 2019).

The dermis is composed of collagen and elastic fibers, being responsible for the resistance and elasticity and this is where the blood vessels, lymphatics, nerves, hair follicles, sebaceous and sweat glands are located (Torres de Jassai et al., 2007). There is a negative correlation between follicular density and fiber thickness. The main histological characteristic of fine fibers is the larger number of secondary follicles, as it has been widely demonstrated that animals with smaller fiber diameter have a higher number of secondary follicles (Atlee et al., 1997; Moore et al., 2011; Vélez et al., 2016).

The fleece varies according to the body region, with the finer, longer and more dense in the dorsal, costal, sacral, brachial and femoral sub-regions (Tolosa et al., 2003; Grigg et al., 2004; Gerken, 2010). For these reasons, integument of scapular, costal and lateral femoral regions was

evaluated by means of stereology to establish comparisons between Suri and Huacaya and to define which of these regions would produce the best fiber for commercial purposes.

2. Material and methods

Alpaca skin biopsies were collected from the Pacamarca Experimental Fund in Llalli, Melgar, Puno - Peru, from Inca Tops SA. Twelve alpacas of both sexes were used. The animals were divided in groups: three Huacaya light (5–7 months), three Huacaya dark (5–6 months); three Suri light (6–17 months) and three Suri dark (17–19 months). Skin biopsies were uniform randomly sampled with an 8 mm punch in three body regions: scapular, costal and lateral femoral and fixed in 10 % formaldehyde solution. They were later cut into 2 mm thick fragments. The distribution of fragments in these four sets was made also following the pattern of systematic uniform random sampling (SURS).

* Corresponding author.

E-mail address: miglino@usp.br (M.A. Miglino).

<https://doi.org/10.1016/j.smallrumres.2022.106629>

Received 5 May 2021; Received in revised form 21 December 2021; Accepted 21 January 2022

Available online 22 January 2022

0921-4488/© 2022 Elsevier B.V. All rights reserved.

2.1. Stereology

2.1.1. Total volume of the skin (reference volume)

The 2 mm-fragments were embedded in paraffin and 5 µm sections were collected on slides and stained with Masson trichrome. The distance between sections was 100 µm. The total volume of the skin (Vref) was estimated using the Cavalieri Method (Gundersen et al., 1999; Howard and Reed, 2010). The image sections were captured using a digital camera connected to the light microscope (Nikon Eclipse 80i, Japan). The following formula was applied to estimate the total volume of skin (Vref):

$$V_{ref} = \sum pS \times a(p) \times t \times k$$

$\sum pS$: points hitting the whole skin fragment; $a(p)$: area of points (square distance between points), t : section thickness; k : the distance between the sections.

2.1.2. Tissue shrinkage

The 8mm-biopsy-fragments were weighed using a digital precision balance (model JK3202B) and their wet weights were converted into volumes considering a tissue density of 1.06 g/cm³. After the estimation of the total volume of the skin by the Cavalieri method above, the tissue shrinkage was calculated as 15 %.

2.1.3. Total volume of the derma, follicular groups, sebaceous and sweat glands

The total volume of the derma (VD), follicular groups (VGF), sebaceous (VGSe) and sweat glands (VGSu) and the fraction volume were calculated. The total volume was estimated multiplying each fraction volume by the reference volume.

2.2. Statistical analysis

A three-way analysis of variance was carried out including factors of breed, colour, region of the body and their interactions. Significant differences were tested using the Tukey test at 5%.

The effect of the animal was analysed in the model as a random repeated measure within breed*colour, as several measures were made on the same animal in different regions: scapular, costal and lateral femoral. There were 8 degrees of freedom in the error term. Degrees of freedom: Breed: 1, colour: 1, breed*colour: 1, region: 2, Animal within (breed*colour): 8. Correlations and principal component analyses between characteristics were calculated. Analysis was carried out in SAS v9.4 (Cary, North Carolina).

3. Results and discussion

The skin is composed by epiderma and derma, the latter presented sebaceous and sweat glands, as well as follicular hair groups, where are the fibers. Vref and VGF were significantly different for body region (Table 1), with femoral region showing higher values than scapula, but costal was not different from femoral or scapula for VGF. Means in the same line followed by different letters are significantly different using the Tukey test ($P < 0.05$). The mean Vref for femoral, costal and scapular regions were respectively: 5.2 (a) mm³, 4.4 mm³ (ab), 4.2 mm³ (b); mean VGF were respectively, 2.8 mm³ (a), 2.0 mm³ (b), 2.2 mm³ (b). Means in the same line followed by different lower case letters and followed by upper case letters are significantly different using the Tukey test ($P < 0.05$). There were also significant interactions between colour and breed for Vref and VD; The mean Vref for light and dark Huacaya were respectively: 4.2 mm³ and 3.8 mm³ (a) and for light and dark Suri were respectively: 4.7 mm³ and 5.6 mm³ (b). The mean VD were respectively for light and dark Huacaya: 1.3 mm³ and 1.1 mm³ (a) and for light and dark Suri were respectively: 1.7 mm³ and 2.5 mm³ (A) (Bb). Dark coats had significantly lower values in Huacaya than Suri breeds

Table 1

Level of Significance and summary of analysis of variance for skin traits in alpacas.

	Vref	VGF	VD	VGSe	VGSu
Source	Pr > F	Pr > F	Pr > F	Pr > F	Pr > F
Breed	0.00	0.09	<.0001	0.32	0.15
Colour	0.52	0.60	0.12	0.24	0.91
Breed*Colour	0.05	0.62	0.00	0.29	0.75
Region	0.02	0.00	0.30	0.48	0.38
Breed*Region	0.40	0.24	0.70	0.31	0.93
Colour*Region	0.53	0.38	0.89	0.52	0.75
Breed*Colour*Region	0.23	0.37	0.47	0.32	0.52
R ²	0.58	0.52	0.66	0.32	0.21
cv	19.34	21.09	30.06	177.83	47.11
Mean in mm ³	4.6	2.3	1.6	0.055	0.17

R² – coefficient of determination, CV – coefficient of variation; the total volume of skin (Vref), total volume of the derma (VD), follicular groups (VGF), sebaceous (VGSe) and sweat glands (VGSu). The mean volume is expressed in mm³.

for both traits while VD was lower in light than dark coats within the Suri breed. Correlations between Vref and VD were high and medium with VGSu, while others were low (Table 2).

The principal component analysis showed a strong relationship between all traits except VGSe (Fig. 1). More follicular groups and skin volume were seen in the lateral femoral region. In dark skinned animals, more skin volume was seen in the Suri compared to the light-colored animals and Dark Suri produced more derma than light-skinned animals or Huacaya. The comparison between Suri and Huacaya fibre diameter showed that the Suri fibre diameter were coarser than Huacaya. The secondary follicles density and the ratio of secondary to primary follicles in Suri was significantly greater than in Huacaya (McGregor and Butler, 2004; Ferguson et al., 2012). In our study, the dark animals presented more total skin volume. Concerning the fibre diameter, the dark colour was coarser than the light according to Lupton et al. (2006a,b). The light fleeces are finer and has less variation in fibre diameter than dark fleeces as reported by McGregor and Butler (2004). No significant differences among colours was observed by Wuliji et al. (2000).

Close relationship between total volume of the derma (VD), follicular groups (VGF), skin volume (Vref) and sweat glands (VGSu) was observed. This indicates that for fiber production, selection for one of these traits should increase the others. This may be useful for breeders when selecting animals for reproduction in breeding schemes. These parameters were not correlated to the fiber diameter as Moore et al. [14] performed with follicle density and ducts of sweat gland. According to them, finer-fibred animals had a higher total follicle density and more sweat glands.

As reported by Vélez et al. (2016) the fiber quality is influenced by the density of the secondary follicle and the total follicles. An allometric relation between mean fibre diameter (MFD) and fleece-free liveweight in sheep was reported; the MFD is proportional to the changes in the skin surface area due to the changes in animal size. This response, however, might change with individual animal and nutritional environment, so the chronological age was not a determinant of wool fibre diameter (McGregor and Butler, 2016). The nutritional conditions has been mentioned to have implications in fibre diameter. When adult alpacas

Table 2

Correlations between coat traits in Alpacas.

	Vref	VGF	VD	VGSe
VvGF	0.72			
VvD	0.85	0.31		
VvVGSe	0.11	0.14	0.12	
VvGSu	0.56	0.37	0.51	0.10

Total volume of skin (Vref), total volume of the derma (VD), follicular groups (VGF), sebaceous (VGSe) and sweat glands (VGSu).

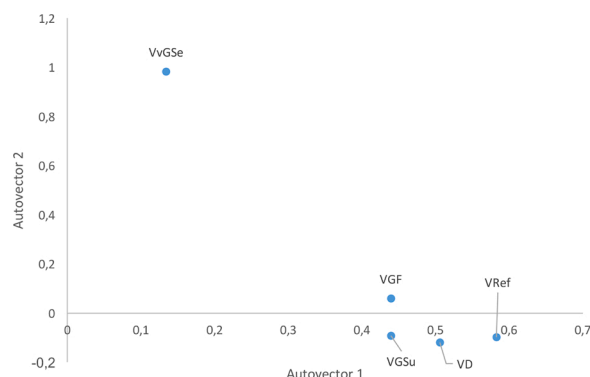


Fig. 1. First two principal components for coat traits in alpacas. Total volume of skin (Vref), total volume of the derma (VD), follicular groups (VGF), sebaceous (VGSe) and sweat glands (VGSu).

lost weight, less fibre grew and consequently the fibre diameter declines (McGregor, 2002). The young alpacas that received 20 % more feed produced more fibre (Lupton et al., 2006a,b). We did not correlate the skin quantification parameters with the age nor nutritional condition, but with colour, breed and corporal region. Our analysis was performed by applying systematic random sampling, based on Stereology, to obtain unbiased representative samples of the alpaca skin. Others methods as the grid sampling has been used for estimating the fleece quality in cashmere goats, in Australian Merino sheep and in alpacas (Butler et al., 1991; McGregor, 1994; Aylan-Parker and McGregor, 2002). It is clear that the accurate sampling and the quantification methods are imperative to obtain reliable data to animal selling programs and breeding.

4. Conclusions

Our study concluded that the femoral region presented higher fiber production. Dark animals had more derma and it was reported close relationship between total skin volume and their fractions volumes: derma, follicular groups and sweat glands.

Authors' contributions

LMDN and MAM conceived and designed the experiment.
LMDN, JB, MK, ADC, ACOC performed the experiment.
LMDN, THCS, SPG, CMM performed the stereological design and analysis.
LMDN, ACOC, THCS, CMM, MAM analysed the data and wrote the paper.

Funding

Coordination for the Improvement of Higher Education Personnel (CAPES) – Brazil.

Availability of data and materials

All analyzed data from this study are included in this manuscript.

Ethics approval

The experiment was approved by the Animal Use Ethics Committee

of the Faculty of Veterinary Medicine and Animal Science of the University of São Paulo (CEUA/FMVZ) under No. 4781170317.

Consent for publication

Not applicable.

Declaration of Competing Interest

The authors declare that they have no competing interest.

References

- Atlee, B.A., Stannard, A.A., Fowler, M.E., Willemse, T., Ihrke, P.J., Olivry, T., 1997. The histology of normal llama skin. *Vet. Dermatol.* 8 (3), 165–176.
- Aylan-Parker, J., McGregor, B.A., 2002. Optimising sampling techniques and estimating sampling variance of fleece quality attributes in alpacas. *Small Rumin. Res.* 44, 53–64.
- Butler, K., Dolling, M., Marland, D.J., Naylor, G.R.S., Phillips, D.G., Veitch, C., 1991. The usefulness of midside sampling as a predictor of top fibre diameter distribution characteristics. In: *Proceedings of the Fibre Diameter Review Conference*. CSIRO, Belmont, Vic., Australia, p. 54.
- Ferguson, M.B., McGregor, B.A., Behrendt, R., 2012. Relationships between skin follicle characteristics and fibre properties of Suri and Huacaya alpacas and Peppin Merino sheep. *Anim. Prod. Sci.* 52, 442–447.
- Gerken, M., 2010. Relationships between integumental characteristics and thermoregulation in South American camelids. *Animal* 4 (9), 1451–1459.
- Grigg, G.C., Beard, L.A., Augee, M.L., 2004. The evolution of endothermy and its diversity in mammals and birds. *Physiol. Biochem. Zool.* 77 (6), 982–997.
- Gundersen, H.J.G., Jensen, E.B.V., Kieu, K., Jensen, J., 1999. The efficiency of systematic sampling in stereology—reconsidered. *J. Microsc.* 193, 199–211.
- Gutiérrez Hermosa, L. Las normas técnicas peruanas y su impacto en el desarrollo de los camélidos peruanos. 2011. http://www.minagri.gob.pe/portal/download/pdf/herramientas/cendoc/videoconferencias/2011/normas_tecnicas_camelidos_13dic11.pdf (accessed 08 Set 2021).
- Howard, C.V., Reed, M.G., 2010. *Unbiased stereology. Three-Dimensional Measurement in Microscopy*, second ed. QTP Publications, Liverpool.
- Lupton, C.J., Elvestad, R.P., Pfeiffer, F.A., Mackinnon, K., 2006a. Effects of age, location, and nutrition on body weight, fiber production, and fiber quality characteristics of penned alpaca males. *J. Anim. Sci.* 84 (1), 58.
- Lupton, C.J., McColl, A., Stobart, R.H., 2006b. Fiber characteristics of the Huacaya Alpaca. *Small Rumin. Res.* 64, 211–224.
- McGregor, B.A., 1994. Measuring cashmere content and quality of fleeces using whole fleece and midside samples and the influence of nutrition on the test method. *Proc. Aust. Soc. Anim. Prod.* 20, 186–189.
- McGregor, B.A., 2002. Comparative productivity and grazing behaviour of Huacaya alpacas and Peppin Merino sheep grazed on annual pastures. *Small Rumin. Res.* 44, 219–232.
- McGregor, B.A., Butler, K.L., 2004. Sources of variation in fibre diameter attributes of Australian alpacas and implications for fleece evaluation and animal selection. *Aust. J. Agric. Res.* 55, 433–442.
- McGregor, B.A., Butler, K.L., 2016. Coarser wool is not a necessary consequence of sheep aging: allometric relationship between fibre diameter and fleece-free liveweight of Saxon Merino sheep. *Animal* 10 (12), 2051–2060.
- Moore, K.E., Blache, D., Maloney, S.K., 2011. Fibre diameter and insulation in alpacas: the biophysical implications. *Small Rumin. Res.* 96 (2–3), 165–172.
- Tolosa, E.M.C., Rodrigues, C.J., Behmer, A.O., Neto, A.G.F., 2003. *Manual de Técnicas para Histología Normal e Patológica*, second ed. Manole, São Paulo.
- Torres de Jasau, J., Vélez, V., Zegar, J., Díaz, G., 2007. Caracterización de la histología de la piel de alpaca. *Sitio Argent Producción Anim.* 1–8.
- Vélez, V., Salazar, J., Pacheco, J., Pezo, D., Franco, F., 2016. Histología cuantitativa de la piel de alpaca diferenciada por calidad de fibra. *Arch Latinoam Prod Anim.* 24 (7), 7–10.
- Wuliji, T., Davis, G.H., Dodds, K.G., Turner, P.R., Andrews, R.N., Bruce, G.D., 2000. Production performance, repeatability and heritability estimates for live weight, fleece weight and fiber characteristics of alpacas in New Zealand. *Small Rumin. Res.* 37, 189–201.